

IAL Chemistry Unit 3 (WCH13) — A* Guide

Practical Skills in Chemistry I · built from all 21 papers and mark schemes, June 2019 – June 2026

1. The paper and what it rewards

50 marks, 1 hour 20 minutes, 4–5 structured questions. No lab work: you describe methods, give observations, process data and evaluate. Most marks are for **exact mark-scheme wording** and **clean calculation chains**. The same ideas recur every session.

Topic	Seen in (of 21)	Typical marks
Titration: colours, results table, mean, multi-step calculation	~17	8–19
Organic preparation: reflux, distillation, separating funnel, drying, % yield	~15	8–18
Flame test method and colours	~14	2–5
Enthalpy experiments: calorimetry, extrapolation, errors	~13	8–13
Inorganic ion tests (halide, sulfate, carbonate, NH_4^+ , nitrate)	~12	4–11
Organic functional group tests	~12	2–8
Graphs: plotting, best fit, extrapolation, tangents	~12	2–5
Uncertainty, % error, improvements	~10	1–4
Gas volumes, molar volume, identifying a metal	~7	3–4
Hazard symbols and precautions	most papers	1–2
IR and mass spectra (short parts)	~7	1–3

2. Mark-scheme rules that decide A* vs A

- **Name and formula:** if you give both, both must be correct. Give one unless asked for both.
- **Ions, not elements:** Ba^{2+} not Ba; iodide / I^- not iodine; "bromide ion" not "bromine ion". Halogen $\neq$ halide.
- **State the state:** "white *precipitate*", not "white" or "white solution". "Steamy fumes" $\neq$ "white smoke". "Clear" is ignored — write **colourless**.
- **Contradictions cancel:** extra wrong observations (e.g. "white ppt and bubbles") lose the mark. List principle applies to multiple tests.
- **Test + result:** the result mark depends on a correct test (or near miss). Always give both.
- **Colours:** use the exact words in Section 3. Methyl orange end-point is **orange**, never red. Phenolphthalein is **pink**, never purple or red.
- **Calculations:** show every step (transferred errors are credited). Do not round intermediate values to 1 SF. Final answer to 2–3 SF unless told otherwise; never 1 SF.

- **Enthalpy answers:** sign + value + kJ mol^{-1} . " kJ/mol^{-1} " is penalised.
- **Percentage uncertainty:** the one place where rounding is strictly checked ($0.8547 \rightarrow 0.85$, not 0.86).
- **Vague words score zero:** "more accurate", "to ensure accuracy", "impurities", "to prevent bumping" (without why), "safety" alone, "wear goggles/lab coat" for a specific hazard.
- **Structures:** -OH drawn horizontally must connect O to C (HO-C on the left). Aldehyde is -CHO, never -COH.
- **Reagent names:** "potassium dichromate(VI)" (not dichromate(IV)), "acidified with sulfuric acid" (never HCl or HNO_3), "phosphorus(V) chloride / PCl_5 " (not PCl_3 as first choice).

3. Inorganic tests — accepted wording

3.1 Flame test (4-5 mark method, asked almost every year)

1. Use a **nichrome or platinum wire** (or silica rod). Not "nickel" alone, not an inoculating loop. Reason: inert / gives no flame colour / high melting temperature.
2. Clean the wire by dipping in **concentrated hydrochloric acid** (only HCl — metal chlorides are more volatile).
3. Dip the moistened wire into the **solid** (paste) and hold it in a **hot, non-luminous (blue, roaring, air-hole open) Bunsen flame**. Never "burn the sample".
4. Observe the colour.

Ion	Write	Avoid
Li^+	red (carmine)	brick-red, orange; "crimson" was rejected in 2026 — write red
Na^+	yellow (orange / gold allowed)	orange-red
K^+	lilac	purple, pink, violet
Rb^+	red-violet / red-purple	brick-red, orange-red
Ca^{2+}	orange-red / brick-red	"red" or "orange" alone
Sr^{2+}	red / crimson (scarlet allowed)	orange
Ba^{2+}	(apple) green	blue-green
Mg^{2+} , Be^{2+}	no colour	white

"Improve the method" answers (Oct 2020): HCl not HNO_3 (chlorides more volatile); nichrome not copper wire (copper gives its own blue-green colour); air-hole open (luminous yellow flame masks colours). **Wooden splint:** cheap and disposable, but wood burns with a yellow flame that masks the colour.

3.2 Anions

Ion	Test	Positive result	Key "why"
CO_3^{2-}	add dilute acid; bubble gas into limewater	effervescence; limewater turns cloudy / milky	$\text{CO}_3^{2-} + 2\text{H}^+ \rightarrow \text{CO}_2 + \text{H}_2\text{O}$
SO_4^{2-}	add dilute HCl (or HNO_3), then $\text{BaCl}_2(\text{aq})$ (or $\text{Ba}(\text{NO}_3)_2$)	white precipitate	acid removes $\text{CO}_3^{2-} / \text{SO}_3^{2-}$, which also give white ppts. Never H_2SO_4 (adds sulfate). $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$
Cl^-	add dilute HNO_3 , then $\text{AgNO}_3(\text{aq})$; then ammonia	white ppt, dissolves in dilute NH_3	HNO_3 removes CO_3^{2-} (Ag_2CO_3 would precipitate). Not HCl (adds Cl^-).
Br^-		cream ppt, dissolves only in concentrated NH_3	
I^-		yellow ppt, insoluble in conc. NH_3	

Distinguishing a halide mixture: add dilute NH_3 — only AgCl dissolves; then conc. NH_3 — AgBr dissolves, AgI remains. A white + yellow mixture becomes "more yellow" after dilute NH_3 because the white AgCl dissolves.

Solid + conc. H_2SO_4	Observations	Products
NaCl	steamy (misty) fumes only	HCl
NaBr	steamy fumes + brown / red-brown fumes	HBr , Br_2 , SO_2
NaI	steamy fumes, purple vapour , black/grey solid, yellow solid, bad-egg smell	HI , I_2 , S , H_2S

Do in a **fume cupboard**: Br_2 , SO_2 , H_2S , HBr , HI are toxic. The orange colour when KBr meets H_2SO_4 in a prep is Br_2 , from oxidation of bromide by sulfuric acid.

Halogen displacement: add $\text{Cl}_2(\text{aq})$ or $\text{Br}_2(\text{aq})$, then hexane, shake. Upper organic layer purple $\rightarrow \text{I}_2$ (iodide present); orange $\rightarrow \text{Br}_2$. $2\text{I}^-(\text{aq}) + \text{Br}_2(\text{aq}) \rightarrow \text{I}_2(\text{aq}) + 2\text{Br}^-(\text{aq})$. Bromine water + NaCl : no reaction (only paler from dilution).

3.3 Cations, nitrates and gases

- NH_4^+ :** add $\text{NaOH}(\text{aq})$ and **warm**; gas turns **damp red litmus blue** (or white smoke with HCl on a glass rod). Paper held at the mouth, not dipped in the solution.
- Group 2 cations:** add a soluble sulfate $\rightarrow$ white ppt with $\text{Ba}^{2+}/\text{Sr}^{2+}$; none with Mg^{2+} or Li^+ . $\text{NaOH}(\text{aq}) \rightarrow$ white ppt with Mg^{2+} , no change with Ba^{2+} .

- **Nitrate decomposition:** Group 2 and Li nitrates give **brown gas** (NO_2) and O_2 (relights a glowing splint). $2\text{Ba}(\text{NO}_3)_2 \rightarrow 2\text{BaO} + 4\text{NO}_2 + \text{O}_2$. Na/K nitrates give nitrite + O_2 only. Complete when no more brown gas / heated to constant mass.
- **Group 2 oxide + water:** solid dissolves, gets warm; forms $\text{M}(\text{OH})_2$. **Ca + water:** effervescence, solid disappears, cloudy white ppt ($\text{Ca}(\text{OH})_2$). $\text{Ca}(\text{OH})_2$ is less soluble when hot (dissolving is exothermic) $\rightarrow$ warming makes it cloudier.
- **Carbonate thermal stability method:** same moles of each, test tube clamped at fixed height in same flame, delivery tube into limewater, time until cloudiness first appears. Some (e.g. BaCO_3) need temperatures a Bunsen cannot reach.

Gas	Test $\rightarrow$ result
H_2	lighted splint $\rightarrow$ squeaky pop
O_2	glowing splint $\rightarrow$ relights
CO_2	limewater $\rightarrow$ cloudy / white ppt
NH_3	damp red litmus $\rightarrow$ blue; white smoke with $\text{HCl}(\text{g})$
HCl	steamy fumes in air; damp blue litmus $\rightarrow$ red; white smoke with NH_3
H_2O	anhydrous CuSO_4 white $\rightarrow$ blue; cobalt chloride paper blue $\rightarrow$ pink (not red)

4. Organic tests

Group	Reagent	Positive result (exact)	Notes
$\text{C}=\text{C}$ (alkene)	bromine water	orange / yellow / brown $\rightarrow$ colourless	not "red". Alternative: acidified KMnO_4 , purple $\rightarrow$ colourless. Alkane: stays orange / purple
$-\text{OH}$ (alcohol or acid)	PCl_5 (solid, dry sample)	steamy / misty fumes (HCl)	not "white smoke". Water also reacts, so the sample must be dry. Alternatives: Na $\rightarrow$ effervescence; carboxylic acid + conc. H_2SO_4 , warm $\rightarrow$ fruity smell
1° / 2° alcohol, aldehyde	$\text{K}_2\text{Cr}_2\text{O}_7$ + dilute H_2SO_4 , warm	orange $\rightarrow$ green	3° alcohol: stays orange. Never HCl / HNO_3
Aldehyde	Fehling's / Benedict's, warm	blue solution $\rightarrow$ (brick-)red precipitate	Tollens': silver mirror. Ketone: no change
Carboxylic acid	Na_2CO_3 / $\text{NaHCO}_3(\text{aq})$ or Mg	effervescence (gas turns limewater cloudy)	blue litmus $\rightarrow$ red also accepted
$\text{CH}_3\text{CO}-$ / $\text{CH}_3\text{CH}(\text{OH})-$	I_2 + $\text{NaOH}(\text{aq})$	pale yellow ppt (antiseptic smell)	distinguishes butanone from butanal

Halogenoalkane	warm with ethanolic AgNO ₃	white / cream / yellow ppt	rate C-I > C-Br > C-Cl; tertiary > primary; rate $\propto$ 1/time
Unsaturation / high C:H	burn	smoky, sooty yellow flame	

Oxidation of alcohols — apparatus logic

- Want the **aldehyde**: heat with **distillation** apparatus so the aldehyde distils off as it forms, preventing further oxidation to the carboxylic acid.
- Want the **carboxylic acid** or **ketone**: heat under **reflux** — vapour condenses and returns to the flask, so oxidation is complete. An open tube loses volatile (and flammable) reactants and products.
- Products: propan-1-ol → propanal → propanoic acid; propan-2-ol → propanone; 2-methylpropan-2-ol (tertiary) → no reaction.
- C₄H₁₀O primary alcohols: butan-1-ol and 2-methylpropan-1-ol both give aldehydes, so the aldehyde test cannot tell them apart.

Spectra used in this paper

IR bond	Range / cm ⁻¹	Mass spectrum points
O-H alcohol	3750–3200	M ⁺ at highest m/z = M _r . Fragments must carry + charge (radicals are not detected). Isomers share M ⁺ . m/z 15 CH ₃ ⁺ ; 31 CH ₂ OH ⁺ (primary alcohol); 43 CH ₃ CO ⁺ ; 45 CH ₃ CHOH ⁺ (secondary). 2-hydroxypropanal gives 15 (CH ₃ ⁺), 3-hydroxypropanal gives 31 (CH ₂ OH ⁺). Br compounds: peaks in pairs (⁷⁹ Br/ ⁸¹ Br).
O-H acid (broad)	3300–2500	
C=O	1740–1680	
C=C	1669–1645	
C-H (alkane)	2962–2853	
C-Cl	700–550	

Following a reaction by IR: the alcohol O-H peak (~3350) decreases, or the C-Cl peak (~590) appears.

5. Apparatus, diagrams and "why" answers

5.1 Diagram mark points (3-4 marks)

Reflux	Distillation
- Round-bottomed or pear-shaped flask (never conical) with liquid, and heat (arrow is enough) Vertical condenser with a water jacket; water in at the bottom, out at the top - Open at the top, no gaps, no stopper; flask joined to condenser - Anti-bumping granules	- Round-bottomed flask + heat + anti-bumping granules - Still head; thermometer bulb level with the side arm (not in the liquid) Sloping Liebig condenser; water in at the lower end, out at the upper end - Sealed on the left, open receiver (or vent) on the right - Separate pieces — a "one-piece" drawing loses a mark; a fractionating column is fine

Separating funnel drawing: tap with stem at the bottom, narrow neck that can be stoppered, two labelled layers. Denser liquid at the bottom: bromoethane, 1-bromobutane, 1-bromopropane are **below** water; cyclohexene, 2-chloro-2-methylpropane (0.87 g cm^{-3}), pentane/limonene, cyclohexane are **above**.

Gas collection over water: sealed flask with delivery tube (bung), tube ending under an inverted measuring cylinder full of water in a trough. A gas syringe is better: sealed, finer graduations, CO_2 does not dissolve.

5.2 Why do we...? (one-mark answers that keep coming back)

Step	Answer that scores
Anti-bumping granules	provide sites (surface) for small bubbles to form → smooth boiling. "Prevent bumping" alone is ignored.
Condenser water in at the bottom	jacket fills completely with no air bubbles → more efficient cooling/condensation
Electric heater / water bath, no Bunsen	organic liquids are flammable
Cool / add acid slowly while mixing	reaction is exothermic; volatile reactant/product would evaporate (or mixture boil over)
Grind KBr (pestle and mortar)	larger surface area → faster reaction
Wash with $\text{Na}_2\text{CO}_3/\text{NaHCO}_3$	neutralise remaining acid (HCl , H_2SO_4 , H_3PO_4)
Open the tap / remove stopper while shaking	release pressure (CO_2 or vapour builds up)
Why not NaOH to neutralise	it reacts with (hydrolyses) the halogenoalkane / is a nucleophile

Wash with water	removes water-soluble / ionic impurities (salts, alcohol, Na_2CO_3)
Anhydrous CaCl_2 / MgSO_4 / Na_2SO_4	drying agent (never "dehydrating agent"); liquid turns cloudy → clear ; then decant or filter. Unsuitable: conc. H_2SO_4 , CuSO_4 , NaOH , CaCO_3
Not dry by heating (bromoethane)	product is volatile and would evaporate first
Collect distillate over a range	range must start just below and end just above the b.p., e.g. cyclohexene (83 °C) 80–86 °C; 1-bromobutane (102 °C) 100–104 °C. Do not start or end on the b.p. itself.
Fractionating column instead of still head	better separation; acts like reflux so the higher-boiling reactant returns (longer reaction, higher yield)
Heat flask gently in distillation	so reaction mixture / reactant is not carried over into the product
Check purity	measure boiling temperature and compare with data book; or IR
% yield below 100 %	incomplete reaction, side reactions, transfer losses, product lost in purification
Dropping funnel for acid	controls rate → steady stream of gas (Cl_2)
Pass Cl_2 before heating metal	flush air out so the metal does not form oxide
CaCl_2 tube before metal ($\text{AlCl}_3/\text{FeCl}_3$)	dry the gas; water would react with the product
Ceramic / mineral wool	holds solid in place while letting gas through; inert
Soda lime / NaOH at the end	absorb excess Cl_2 / HCl
FeCl_3 mass higher than expected	product absorbs water from the air (hydrated)
Remove delivery tube before stopping heating	prevent suck-back: tube cools, pressure falls, cold water drawn into the hot tube, which cracks
Let H_2 flow before lighting (CuO reduction)	remove air; H_2 -air mixture is explosive. Cool in H_2 or the copper re-oxidises (turns black)
Clean Mg ribbon with emery paper	remove MgO layer so only Mg is weighed
Polystyrene cup inside a beaker	stops the cup tipping over (stability), not insulation
Excess reagent (Zn, Mg)	ensures all of the other reactant reacts; show it with a mole calculation
Weigh the weighing bottle again after tipping	accounts for solid left behind (mass by difference)
Crystals from solution	heat to evaporate until crystallisation point (not dryness), leave to cool and crystallise, filter, dry between filter papers

Heat to constant mass	weigh crucible, add solid, weigh, heat, cool, reweigh; repeat until mass unchanged → reaction complete
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6. Titration

6.1 Making a standard solution (4-6 marks)

Standard solution = solution of **accurately known concentration**.

1. Calculate mass: $n = c \times V$, $m = n \times M$ (e.g. 100 cm^3 of $0.0500 \text{ mol dm}^{-3}$ $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$, $M = 126 \rightarrow 0.630 \text{ g}$).
2. Weigh accurately; dissolve in a small volume of **distilled/deionised water** in a beaker, stirring.
3. Transfer through a funnel into a **volumetric flask**; rinse beaker and funnel, adding the **washings** to the flask.
4. Make up **to the mark** with distilled water (dropwise at the end, bottom of meniscus on the line, eye level).
5. Stopper and **invert** several times → uniform concentration. Never filter into the flask.

6.2 Procedure points

- Rinse the burette with the solution it will hold (water would dilute it). Rinse the pipette with its solution. Conical flask: distilled water only — extra water does not change moles.
- Use a volumetric **pipette with filler** for 25.0 cm^3 . Fill the burette below eye level (NaOH is an irritant and may splash into eyes); remove the funnel.
- Rough titration first (to find the approximate end-point, saving time). Then add quickly to $1\text{--}2 \text{ cm}^3$ below the rough value, then **dropwise with swirling** until the colour just changes.
- Read the **bottom of the meniscus at eye level**. Record to 2 d.p. ending in 0 or 5 (24.20, not 24.2).
- **Concordant** = within 0.10 cm^3 (0.20 accepted). Mean only concordant titres; exclude the rough. If two titres already agree, a further titration is not needed.
- Dilution by 10 before titrating: otherwise the titre would exceed the burette's capacity.

Indicator	Titrating	Colour change
Methyl orange	acid added to alkali / carbonate	yellow → orange (not red)
Phenolphthalein	alkali added to acid	colourless → (pale) pink
Phenolphthalein	acid added to alkali	pink → colourless
K_2CrO_4 (Ag^+ titration)	AgNO_3 into Cl^-	AgCl precipitates first because it is much less soluble. Remove Ba^{2+} first (as BaSO_4) or BaCrO_4 forms.

Error	Effect on titre	Reason
Burette rinsed with water	increases	titrant diluted
Pipette rinsed with water	decreases	fewer moles in the flask
Conical flask wet	no change	moles unchanged
Pipette blown out (last drop forced in)	increases	more than 25.0 cm ³ delivered
NaOH left open to air (as titrant)	increases	reacts with CO ₂ , less NaOH
Water evaporates from NaOH (as titrant)	decreases	more concentrated

6.3 Calculation chain

mol titrant = titre/1000 × c → use the **equation ratio** → mol in aliquot → × (250/25) for the whole flask → mass (× M) or concentration (÷ V) → % / M_r / n.

- Na₂CO₃ + 2HCl: mol Na₂CO₃ = mol HCl ÷ 2. Ba(OH)₂, Ca(OH)₂, H₂SO₄, diprotic acids: also 1 : 2. Citric acid: 1 : 3.
- Back titration (antacid): mol acid added – mol acid left (from titration × dilution) = mol acid reacted with the tablet.
- Concentration in g dm⁻³ = mol dm⁻³ × M.

Worked — water of crystallisation (Jun 2026). 3.15 g Na₂CO₃·nH₂O in 250 cm³; 25.0 cm³ needs 22.00 cm³ of 0.100 mol dm⁻³ HCl.
 mol HCl = 22.00 × 0.100/1000 = 2.20 × 10⁻³ → mol Na₂CO₃ = 1.10 × 10⁻³ → in 250 cm³ = 1.10 × 10⁻²
 M = 3.15 ÷ 0.0110 = 286 → water = 286 – 106 = 180 → n = 180/18 = **10** (must be a whole number).

Worked — % purity (Jun 2019). Mean titre 24.20 cm³ of 0.095 mol dm⁻³ HCl against 25.0 cm³ of NaOH made from 0.95 g in 250 cm³.
 mol HCl = 2.299 × 10⁻³ = mol NaOH (1:1) → × 10 = 0.02299 → mass = 0.02299 × 40 = 0.9196 g → purity = 0.9196/0.95 × 100 = **96.8 %**. Never give >100 %.

Worked — identify a metal (Jun 2022, MHCO₃). mol HCl = 2.0025 × 10⁻³ = mol MHCO₃ in 25 cm³ → × 10 → M_r = 2.00 ÷ 0.020025 = 99.88 → A_r(M) = 99.88 – 61 = 38.9 → potassium. (For M₂CO₃·10H₂O: A_r = (M_r – 60 – 180)/2.)

7. Enthalpy experiments

7.1 Solution calorimetry method

1. Measure a known volume into a polystyrene cup (in a beaker for stability) with a pipette or burette.
2. Record the temperature every 30 s for ~3 min to get a steady starting value.
3. Add the solid / second solution at a known time (e.g. 3.0 min), stir, and keep recording every 30 s. Do not record at the moment of mixing — temperature is changing and not

uniform.

4. Plot temperature against time. Draw a best-fit line through the cooling (or warming) points and **extrapolate back to the time of mixing**. ΔT = extrapolated value – starting value.

Why extrapolate: the reaction is not instantaneous, so heat is lost while it happens; extrapolation corrects for this.

$Q = m c \Delta T$ (m = mass of **solution** in g = total volume; $c = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$)

$\Delta H = -Q \div (1000 \times \text{mol of limiting reactant}) \text{ kJ mol}^{-1}$; negative for a temperature rise.

Assumptions (1 mark each): density of solution = 1 g cm^{-3} ; specific heat capacity of solution = water; heat capacity of cup/metal/thermometer negligible; mass of solid ignored. Neutralisation is **per mole of water**: with $\text{H}_2\text{SO}_4 + \text{NaOH}$, $\text{mol H}_2\text{O} = \text{mol NaOH}$.

Worked (Jun 2025). 50.0 cm^3 of $1.00 \text{ mol dm}^{-3} \text{ Zn}^{2+}$ with 2.20 g Mg (excess: $0.0905 \text{ mol} > 0.0500 \text{ mol}$). $\Delta T = 55.2 - 19.4 = 35.8 \text{ }^\circ\text{C}$.

$Q = 50.0 \times 4.18 \times 35.8 = 7482 \text{ J} \rightarrow \Delta H = -7.482 \div 0.0500 = \mathbf{-150 \text{ kJ mol}^{-1}}$.

7.2 Variations

- **Endothermic dissolving (NH_4Cl):** extrapolate to the minimum temperature. In a glass beaker, heat enters faster: minimum is higher and temperature recovers faster.
- **Thermometric titration:** plot temperature vs volume added; two straight best-fit lines; intersection gives end-point volume and maximum ΔT . m = total volume of both solutions. Lower ΔT than expected: heat loss, acid added in small portions/slowly, volume increasing.
- **Displacement (Zn/CuSO_4 , Mg/Zn^{2+}):** use metal powder in excess; prove excess with moles.
- **Hess cycle (hydration of CuSO_4):** $\Delta H = \Delta H_1 - \Delta H_2$. Cannot measure directly: hard to add exactly the right amount of water; some solid dissolves while some is not fully hydrated.
- **Glass beaker instead of polystyrene:** value **less exothermic** because glass conducts heat / has a higher heat capacity.

7.3 Combustion (spirit burner)

Measure: mass (volume) of water; initial and final water temperature; mass of burner before and after. Keep a lid (cap) on the burner when not lit and reweigh at once (fuel evaporates).

Worked (Jan 2023). 150 g water, $\Delta T = 7.3 \text{ }^\circ\text{C}$, 0.32 g ethanol. $Q = 150 \times 4.18 \times 7.3 = 4577 \text{ J}$; $\text{mol} = 0.32/46 = 6.957 \times 10^{-3}$; $\Delta H = -4.577/0.006957 = \mathbf{-658 \text{ kJ mol}^{-1}}$.

- **Why much less exothermic than the data book:** heat lost to surroundings and apparatus; incomplete combustion (soot); fuel evaporates from the wick; non-standard conditions.
- **Burner lit before weighing / left burning:** mass burned is overstated $\rightarrow$ value less exothermic.

- **Improvements:** draught shield, lid on the water container, copper can (conducts), stir water, shorter distance to flame. A copper coil carrying flue gases through the water reduces heat loss; a steady air supply gives complete combustion.

7.4 Improvement answers that score

Lid on the cup (less heat loss); record temperature more often / use a data logger (more precise extrapolation); thermometer with finer divisions or digital (more precise ΔT); burette/pipette for volumes (lower uncertainty); add the reagent in one portion, quickly. Not credited: "repeat the experiment", "more accurate balance", "better insulation" with no detail.

8. Uncertainty, error and evaluation

% uncertainty = (uncertainty per reading $\times$ number of readings $\div$ measured value) $\times$ 100

- Titre (two burette readings): $\pm 0.05 \times 2 = \pm 0.10 \text{ cm}^3$. E.g. $0.10/11.70 \times 100 = 0.85 \%$.
- Temperature change (two readings): $\pm 0.1 \times 2$. E.g. $0.2/7.8 \times 100 = 2.56 \%$.
- Mass by difference (two weighings): $2 \times 0.0025/0.030 \times 100 = 16.7 \%$.
- Pipette / volumetric flask: one reading ($0.04/25 = 0.16 \%$; $0.15/250 = 0.06 \%$).

- **% error** = |experimental – data book| $\div$ **data book** $\times$ 100 (divide by the data value, not yours).
- **Is the result consistent?** Find the range: experimental $\pm$ (total % uncertainty $\times$ experimental value). If the data value lies inside, the difference is explained by measurement uncertainty; if not, there is a systematic error (heat loss, gas escape).
- **Uncertainty vs accuracy:** uncertainty gives the range in which the true value lies; accuracy is how close the result is to the true (data book) value.
- **Reduce % uncertainty:** measure a bigger quantity — larger titre (more dilute titrant or more sample), larger ΔT (more reagent or smaller volume of solution), larger mass. Check limits: gas volume must not exceed the cylinder/burette, titre must stay below 50 cm^3 .
- **Thermometer reading only to $\pm 1^\circ \text{C}$** cannot show a small change — resolution too low.
- **Anomaly:** if all but one point lie on the line, repeating is unnecessary — the trend is clear and the anomaly is excluded from the best-fit line.

9. Gases, yields and other calculations

9.1 Molar gas volume ($24.0 \text{ dm}^3 \text{ mol}^{-1} = 24\,000 \text{ cm}^3$ at RTP)

Identify a Group 2 carbonate (Jun 2024). 0.320 g MCO_3 gives $89.0 \text{ cm}^3 \text{ CO}_2$. $\text{mol} = 89.0/24\,000 = 3.708 \times 10^{-3}$; $M_r = 0.320/0.003708 = 86.3$; $A_r = 86.3 - 60 = 26 \rightarrow$ **magnesium** (nearest Group 2 element).

Measure molar volume (Jun 2023). $0.126 \text{ g Ca} \rightarrow 72.0 \text{ cm}^3 \text{ H}_2$. $\text{mol Ca} = 0.126/40.1 = 3.142 \times 10^{-3}$; $V_m = 72.0/0.003142 = \mathbf{22.9 \text{ dm}^3 \text{ mol}^{-1}}$ (units required).

- **Why too low:** gas escapes before the bung is inserted; CO_2 dissolves in water; metal partly oxidised (impure). Using ethanoic acid (weak) instead of HCl slows the reaction so less gas escapes.
- **Gas M_r from a lighter:** $M_r = \text{mass lost} \div (V/24\,000)$. Level the water inside and outside the cylinder so the gas is at atmospheric pressure. If gas is lost, mass loss seems smaller $\rightarrow M_r$ too small.
- **Mg in a burette (Jan 2025):** nothing happens at first because the acid has not yet diffused down to the Mg.

9.2 Percentage yield

Liquid reactant: mass = volume $\times$ density $\rightarrow$ mol reactant $\rightarrow$ mol product (equation ratio) $\rightarrow$ theoretical mass $\rightarrow$ % yield = actual $\div$ theoretical $\times 100$. Dividing masses directly (product g $\div$ reactant g) scores **zero**.

Worked (Jun 2026). 10.0 cm^3 propan-1-ol (0.80 g cm^{-3}) $\rightarrow 8.00 \text{ g} \rightarrow 0.1333 \text{ mol}$. Theoretical 1-bromopropane = $0.1333 \times 122.9 = 16.39 \text{ g}$. Yield = $7.98/16.39 \times 100 = \mathbf{48.7 \%}$.

9.3 Other recurring calculations

- **Empirical formula from combustion:** masses of H_2O (silica gel tube) and CO_2 (soda-lime tube) — weigh each tube before and after. Use pure O_2 , not air (air contains CO_2 and H_2O ; ensures complete combustion). Mass O = sample — C — H.
- **Formula of copper oxide:** moles Cu and O from masses; 1:1 $\rightarrow \text{CuO}$. If copper re-oxidises on cooling, it looks as if less oxygen was lost.
- **Mixture analysis:** e.g. $\text{KHCO}_3/\text{K}_2\text{CO}_3$: mass lost = $\text{H}_2\text{O} + \text{CO}_2$ (62 g per 2 mol KHCO_3) $\rightarrow$ mol $\text{KHCO}_3 \rightarrow$ mass $\rightarrow$ remainder is K_2CO_3 . Small masses $\rightarrow$ larger % weighing uncertainty.
- **Precipitate from mixture:** mass $\text{AgI} \rightarrow$ mol $\text{AgI} \rightarrow$ mol $\text{CaI}_2 (\div 2) \rightarrow$ mass $\rightarrow$ % of other component.
- **Precipitate-height graph:** two straight lines; the break point = volume where reactant stops being in excess; then mol ratio gives M_r (e.g. $\text{Pb}(\text{NO}_3)_2$, A_r 207).

10. Graphs and rates

- Independent variable on x. Labels with units: "Time / s", "Volume / cm^3 ", "Temperature / $^\circ\text{C}$ ". Use linear scales — a non-uniform scale loses both plotting marks.
- Points must cover **at least half the grid in both directions**; plot to within half a small square.
- Best-fit straight line or smooth curve; ignore the anomaly; only go through the origin if justified.
- Read values with construction lines shown on the graph.

- Rate at a point: draw a **tangent**, gradient = $\Delta y/\Delta x$, give units ($\text{cm}^3 \text{s}^{-1}$). Rate from time: rate $\propto 1/t$, units s^{-1} (e.g. $1/33 = 0.030 \text{s}^{-1}$).
- Rate increases with temperature as an upward curve. An initial slow part may be due to catalyst not yet mixed (induction) or the mixture warming up (exothermic).
- Disappearing-cross reaction too fast: reduce the reactant concentration.

11. Hazards and safety

Symbol	Meaning	Precaution that scores
Flame	flammable	no naked flames; electric heater / water bath
Flame over circle	oxidising	keep away from flammable materials
Skull and crossbones	toxic	fume cupboard (for gases: Cl_2 , Br_2 , NO_2 , SO_2 , HCl)
Corrosion (liquid dripping onto hand and surface)	corrosive	gloves; dilute; use a tray
Exclamation mark	harmful / irritant	gloves; avoid splashes into eyes
Person with star on chest	serious health hazard	fume cupboard; avoid inhalation
Dead tree and fish	harmful to environment	organic waste bottle, not down the sink

Precautions must match the hazard. Goggles, lab coat or "be careful" score nothing. Excess Cl_2 can be absorbed in NaOH : $2\text{NaOH} + \text{Cl}_2 \rightarrow \text{NaCl} + \text{NaClO} + \text{H}_2\text{O}$.

12. Final checklist before the exam

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| <ul style="list-style-type: none"> • Every observation: colour + state (precipitate, solution, fumes, gas). • Every test: reagent + condition (warm, acidified, dry) + result. • Colour changes: both colours, in the right order ("from ... to ..."). • Methyl orange $\rightarrow$ orange; phenolphthalein $\rightarrow$ pink / colourless. • Sulfate test: $\text{HCl} + \text{BaCl}_2$. Halide test: $\text{HNO}_3 + \text{AgNO}_3$. • Flame test: nichrome, conc. HCl, non-luminous flame, K^+ lilac. • Titres to 2 d.p.; mean of concordant only; rough excluded. | <ul style="list-style-type: none"> • Mole ratio from the equation; $\times 10$ for 250 cm^3 from 25.0 cm^3. • $Q = mc\Delta T$ with mass of solution; ΔH negative for exothermic; kJ mol^{-1}. • % uncertainty: two readings $\rightarrow$ double; round correctly. • % yield via moles, using density for liquids. • Diagrams: no gaps, no sealed system, correct water flow, flask not conical. • Drying agent, not dehydrating agent; cloudy $\rightarrow$ clear. • Specific hazard $\rightarrow$ specific precaution. • Never give a % above 100 or an n that isn't a whole number. |
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Appendix — what each paper tested

Use this to pick papers for targeted practice.

Session	Main content
Jun 2019	$(\text{NH}_4)_2\text{CO}_3$ decomposition gas tests; alcohol oxidation apparatus; dehydration over Al_2O_3 (suck-back); thermometric displacement $\rightarrow \text{A}_\text{r}$; NaOH purity titration
Oct 2019	NaNO_3 /ions; propanoic acid, propanol, propanal; spirit-burner combustion; $\text{CuO} + \text{H}_2$ formula; diprotic acid M_r ; limonene extraction
Jan 2020	Halide/bromine water, NH_4^+ , dichromate, hexane/hexene; $\text{Mg} + \text{HCl}$ enthalpy; Na_2CO_3 purity; bromoethane prep
Oct 2020	$\text{Ba}(\text{NO}_3)_2$ decomposition; identifying unlabelled solutions; flame-test improvements; alcohol/acid IR; neutralisation enthalpy; $\text{M}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ titration, crystallisation
Jan 2021	Identifying solutions (ionic equation); ethanedioic acid standard + NaOH; MCO_3 gas volume + Hess; chloroalkane prep, % yield, hydrolysis rates
Jun 2021	Flame test; sulfate/carbonate; alkene/alcohol tests, IR, MS; $\text{Ba}(\text{OH})_2$ titration; 1-bromobutane prep and yield
Oct 2021	Li salt and nitrate; geraniol tests, separation, hazards; hydration of CuSO_4 (extrapolation); disappearing-cross rate; NaOH titration uncertainty
Jan 2022	NH_4Cl dissolving (endothermic graph); combustion analysis + IR/MS; $\text{BaCl}_2 \cdot x\text{H}_2\text{O}$ by Ag^+ titration; AlCl_3 preparation
Jun 2022	$\text{Zn} + \text{CuSO}_4$ extrapolation; MHCO_3 titration, uncertainty range; cyclohexene preparation
Oct 2022	Sulfate test; flame test; halides; antacid back titration; precipitate-height graph (PbI_2); alcohol oxidation and product tests
Jan 2023	KI/bromine, layers; ethanol combustion and uncertainty; citric acid descaler titration; C_4 alcohol isomers
Jun 2023	BaI_2 , $(\text{NH}_4)_2\text{SO}_4$; hydroxypropanal MS; butanoic acid by reflux; $\text{Ca} + \text{water}$ molar volume; $\text{Ca}(\text{OH})_2$ titration
Oct 2023	Flame test; separating $\text{Cl}^-/\text{Br}^-/\text{I}^-$; CaCO_3 gas-volume graph $\rightarrow V_\text{m}$; cyclohexene prep and yield; standardising HCl
Jan 2024	Functional-group tests; citric acid in lemon juice; H_2O_2 /iodide, solvent extraction; density graph; ethanol combustion (copper coil)
Jun 2024	CuCO_3 molar volume, MCO_3 identity; bromoethane prep; ICl/ICl_3 equilibrium; $\text{H}_2\text{SO}_4/\text{NaOH}$ neutralisation
Oct 2024	Group 2 flame colours, carbonate stability; thermometric titration; reflux product identification; H_2O_2 foam rate (tangent)
Jan 2025	Flame test, NH_4^+ , $\text{Ag}_2\text{CO}_3/\text{AgBr}$; Mg molar volume in burette; ethanedioic standard + $\text{Ba}(\text{OH})_2$; 2-chloro-2-methylpropane

Jun 2025	Propanol isomers by MS; Rb/K/Sr/Li, halides; ion migration (Ag_2CrO_4); FeCl_3 preparation; Mg/Zn^{2+} enthalpy
Oct 2025	Alcohol isomers; $\text{Ca}^{2+}/\text{Ba}^{2+}$, I^-/AgBr ; $\text{Na}_2\text{CO}_3 \cdot n\text{H}_2\text{O}$; $\text{KHCO}_3/\text{K}_2\text{CO}_3$ mixture; 2-chloro-2-methylpropane (faulty apparatus)
Jan 2026	$\text{CaI}_2/\text{CaCl}_2$ mixture; thermometric NaOH, rinsing effects, flask vs pipette uncertainty; propanoic acid; gas M_r
Jun 2026	Flame Li/K; conc. H_2SO_4 + halides; NH_4^+ , sulfate; hydroxypropanal; Na_2CO_3 standard, n, constant mass; 1-bromopropane